

## Exercises 4.5

1) a)  $n^{\lg \lg 4^2} = n^{1/2}$  and  $1 \neq n^{1/2 - \epsilon}$   
 $\forall$  positive  $\epsilon < 1/2$  therefore case 1 applies and  $T(n) = \Theta(n^{1/2})$ .

b)  $n^{\lg \lg 4^2} = n^{1/2}$  and  $\sqrt{n} = \Theta(n^{1/2} \lg^0 n)$   
So case 2 applies and  $T(n) = \Theta(n^{1/2} \lg n)$ .

c)  $n^{\lg \lg 4^2} = n^{1/2}$  and  $\sqrt{n} \lg^2 n = \Theta(n^{1/2} \lg^2 n)$   
So case 2 applies and we have  $T(n) = \Theta(n^{1/2} \lg^3 n)$ .

d)  $n^{\lg \lg 4^2} = n^{1/2}$  and  $n = \Omega(n^{1/2 + \epsilon})$   
 $\forall$  positive  $\epsilon \geq 1/2$ .

Regularity condition is also satisfied because  $a f\left(\frac{n}{b}\right) = \frac{2n}{4}$

$$= \frac{n}{2}$$

And  $\frac{n}{2} \leq c f(n)$

$$= cn$$

$\forall c$  such that  $0.5 \leq c \leq 1$

Therefore  $T(n) = \Theta(n)$ .

e)  $n^{\lg \lg 4^2} = n^{1/2}$  and  $n^2 = \Omega(n^{1/2 + \epsilon})$   
 $\forall$  positive  $\epsilon \leq 3/2$ .

Regularity condition is also satisfied because  $a f\left(\frac{n}{b}\right) = 2\left(\frac{n}{4}\right)^2$   
 $= \frac{2n^2}{16}$   
 $= \frac{n^2}{8}$

And  $n^2/8 \leq c f(n)$   
 $\leq cn^2 \quad \forall c \text{ such that}$   
 $1/8 \leq c < 1$

Thus case 3 applies so  $T(n) = \Theta(n^2)$

2)  $T(n) = aT(n/4) + \Theta(n^2)$

The watershed function  $n^{\log_4 a}$  should be less than  $n^{\lg 7}$ :-

$$\begin{aligned} n^{\log_4 a} &< n^{\lg 7} \\ \log_4 a &< \lg 7 \\ \frac{\lg a}{\lg 4} &< \lg 7 \\ \lg a &< \lg 49 \\ a &< 49 \end{aligned}$$

So the <sup>largest</sup> ~~smallest~~ integer value for  $a$  is 48 and the running time would be  $\Theta(n^{\log_4 48})$



3)  $T(n) = T(n/2) + \Theta(1)$

$n^{\lg 1} = n^0 = 1$  and since the  
 $\forall f(n) \in \Theta(1)$ ,  $f(n) = \Theta(1 \lg^0 n)$ ,  
 Case 2 applies and we have  
 $T(n) = \Theta(\lg n)$

4) Suppose  $a f(n/b) \leq c f(n)$  with  $f(n) = \lg n$ ,  
 $a=1$  and  $b=2$ . Then we have:-  
 and all sufficiently large  $n$

$\cancel{1} \lg(n/2) \leq \cancel{c} \lg n$   
 $\lg n - \lg 1 \leq \lg n$   
 $\lg n - \lg n \leq 1$   
 $\lg n (1 - c) \leq 1$   
 $\lg n \leq 1 / (1 - c)$   
 $n \leq 2^{1/(1-c)}$

So the bound holds only up to  
 some finite value of  $n$ , thus  
 Contradicting our supposition that  
 it must hold for all sufficiently  
 large values of  $n$ , therefore the  
 regularity condition does not hold.

5) ~~Let  $b=2$ , and  $a=1$ , and  $\epsilon = \lg 2 = 1$ ,  
 then  $f(n) = 2^{\lg n}$~~

Suppose  $\exists$  an  $\epsilon > 0$  such that  
 $f(n)$  i.e.  $\lg n = \Omega(n^{\lg^{1+\epsilon}})$   
 $\lg n = \Omega(n^\epsilon)$

But  $n^\epsilon$  is a positive power  
of  $n$  so  $\lg n = \Omega(n^\epsilon) = \omega(\lg n)$   
the which contradicts our supposition.

5) We must have  $f(n) = \Omega(n^{\lg_b a + \epsilon})$ ,  
so for some  $\epsilon > 0$ , so:-

$$2^{\lceil \lg n \rceil} = \Omega(n^{\lg_b a + \epsilon})$$

Since  $2^{\lceil \lg n \rceil} = \Theta(n)$  i.e.  $2^{\lceil \lg n \rceil} = \Omega(n)$ ,  
we must have  $\lg_b a + \epsilon \leq 1$ . So  
we must have  $\lg_b a < 1$ , therefore  
we must have  $a < b$ .

For the regularity condition to fail,  
we must show that  $\forall c < 1$  and  
 $\nexists \forall n_0 > 0, \exists n \geq n_0$  such that  
 $a f(n/b) > c f(n)$  So we have i.e.  
 $a 2^{\lceil \lg n/b \rceil} > c 2^{\lceil \lg n \rceil}$

~~Let  $a=2$~~

Let  $a=2$  and  $b=3$ , then we  
have:-

$$2^{\lceil \lg n - \lg 3 + 1 \rceil} > c 2^{\lceil \lg n \rceil}$$

eg. Whenever  $1 - \lg 3 \approx -0.585$ ,  
so whenever  $n = 2^k$  if  $n$  is a  
power of 2,  $\lceil \lg n - \lg 3 + 1 \rceil = \lceil \lg n \rceil$   
and so  $2^{\lceil \lg n - \lg 3 + 1 \rceil} = 2^{\lceil \lg n \rceil}$ .



Since  $c < 1$ , it follows that whenever  $n$  is a power of 2,   
 $\cancel{2} \times 2^{\lceil \lg n/3 \rceil} = 2^{\lceil \lg n - \lg 3 + 1 \rceil} > c \cdot 2^{\lceil \lg n \rceil}$ ,   
 therefore the regularity condition fails.   
 For  $a = \cancel{4} \cancel{7} 2$  and  $b = \cancel{2} 3$ , choose any  $\epsilon \leq 1 - \log_3 2$  to satisfy   
 the first condition i.e.  $2^{\lceil \lg n \rceil} = \Omega(n^{\log_3 a + \epsilon})$